

Project Design Phase Proposed Solution

Date	01 October 2026
Team ID	6ab7f31f8c7c66e70672939b
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Students often find it difficult to understand complex academic concepts when explanations are too technical, too lengthy or not suited to their level of understanding, and they may not get their doubts clarified quickly. After understanding one topic, they may not know which prerequisite or related concept to learn next. They also spend considerable time reviewing long learning content and may not have a simple way to practise and check their understanding. Learning support is usually spread across different resources, so students have to switch between them and look for suitable explanations, summaries, practice questions and next topics on their own. There is therefore a need for an accessible AI-powered learning assistant that brings these forms of support together in one simple interface. Core student problems addressed • Problem 1: Difficulty understanding complex academic concepts. • Problem 2: Difficulty following a structured learning progression. • Problem 3: Difficulty efficiently reviewing and practising learning content.
2.	Idea / Solution description	EduGenie is a unified AI-powered learning assistant that supports multiple stages of a student's learning process through a simple web interface. It is designed as one connected learning-support experience rather than five separate tools. Step 1 – Student Input The student opens the EduGenie web interface and enters a learning request: a difficult concept, an academic question, a long learning passage, a topic for a quiz, or a topic or learning context for recommendations. Step 2 – Appropriate Learning Module EduGenie routes the request to the matching function: Concept Explanation (/explain), Academic Q&A (/qa), Summarization (/summarize), Quiz (/quiz) or Learning Recommendations (/learn/recommendations). Step 3 – AI Processing • LaMini-Flan-T5-783M: simplified concept explanation. • Google Gemini 3.8 Flash (through the Google Gemini API): Q&A, quiz generation, summarization and learning recommendations. Step 4 – Learning Output The generated result is returned through the web application as a simplified explanation, an answer or clarification, a concise summary, quiz questions (multiple-choice with options), or recommended related

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		concepts and next learning steps, which may progress from beginner towards advanced topics and may include learning resources. Step 5 – Continued Learning The student uses the output to understand a concept, review content, practise with a quiz and continue to the next topic. How the solution addresses the problems • Problem 1: simplified explanation and academic Q&A. • Problem 2: learning recommendations. • Problem 3: summarization and quiz generation. The interface is built with HTML and CSS, rendered with Jinja2 templates, and served by a FastAPI (Python 3.10+) backend running on Uvicorn. EduGenie is intended to support students' learning, not to replace teachers or formal education, and AI-generated output may not always be correct, so students should verify important information.
3.	Novelty / Uniqueness	The novelty of EduGenie lies mainly in how existing AI capabilities are combined and used for student learning support, not in claiming that the individual AI techniques are new inventions. • Integrated learning assistance: explanation, Q&A, summarization, quizzes and learning recommendations are brought together in one lightweight interface. • Model-task alignment: LaMini-Flan-T5-783M is used for simplified concept explanation, while Google Gemini 3.8 Flash is used for the broader generative learning tasks. • Lightweight architecture: a simple HTML/CSS frontend with Jinja2, FastAPI, Python and Uvicorn keeps the prototype relatively lightweight. • Learning continuity: the functions together support the sequence Understand → Clarify → Review → Practice → Continue Learning.
4.	Social Impact / Customer Satisfaction	The primary user is the student / learner. The following impacts are intended outcomes; no satisfaction or performance measurements have been collected. • Improved understanding: simplified explanations can help students approach difficult concepts more easily. • Faster doubt clarification: students can ask academic questions and receive AI-generated responses without waiting. • Efficient revision: summaries can reduce the amount of content students need to go through while revising. • Active practice: generated quizzes give students a chance to check their understanding. • Guided learning: recommendations can help students identify relevant concepts and possible next steps. • Accessibility: the lightweight web-based design can be used without a complex application setup.

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5.	Business Model (Revenue Model)	Current status EduGenie is an academic prototype. No commercial revenue model is currently implemented and no revenue has been demonstrated. Potential future revenue models (not currently implemented) • Freemium model: basic learning assistance at no cost, with advanced usage or additional capabilities in a paid tier. • Educational institution licensing: schools, colleges and training institutes could license the platform for student learning support. • Institutional subscription: organizations could pay a recurring fee based on usage or number of learners. • Platform integration: the system could be integrated into existing educational platforms through suitable APIs or service agreements.
6.	Scalability of the Solution	Current prototype EduGenie currently runs as a lightweight application in a local / development environment. Its separation into a frontend, a backend and individual modules and endpoints for explanation (/explain), Q&A (/qa), quiz (/quiz), summarization (/summarize) and learning recommendations (/learn/recommendations) provides a foundation for future extension. The prototype is not claimed to be horizontally scalable or highly available. Future scalability (possible, not currently implemented) If deployed as a production service, EduGenie could be expanded through production server deployment, multiple application instances, load balancing, health checks and monitoring, sufficient external AI service capacity, and modular addition of new features to support more users. Possible future expansion Multilingual learning, mobile access, voice interaction, adaptive learning, progress tracking and integration with educational institutions are possible future enhancements. None of these is part of the current solution.